Zomato Order & Restaurant Analysis Using Power BI

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Task 1: Upload Dataset into MySQL

Create two tables:

- Zomato_Restaurants
- Zomato_Orders

Import the **Zomato_Orders.csv** and **Zomato_Restaurants.csv** files into MySQL.

Query | Query History

```sql
--Create two tables & Import the Zomato_Orders.csv and Zomato_Restaurants.csv files into MySQL.


create table zomato_restuarants(restaurant_id varchar primary key,restaurant_name varchar,city text,
area varchar,cuisine text,avg_rating real,total_ratings real,price_range text,
delivery_available text);

create table zomato_orders(order_id varchar,restaurant_id varchar references zomato_restuarants(restaurant_id)
,customer_id varchar,order_date date,order_time time,delivery_time int,total_cost real,item_count int,
payment_method text,customer_rating real);
```

Task 2 :Perform Simple Data Transformations Using SQL

```sql
--Handle NULL values (if any) by replacing them with appropriate values
select COALESCE(restaurant_id::TEXT, 'No id') AS restaurant_id,
    COALESCE(restaurant_name::TEXT, 'No name') AS restaurant_name,
    COALESCE(city::TEXT, 'No city') AS city,
    COALESCE(area::TEXT, 'No area') AS area,
    COALESCE(cuisine::TEXT, 'No cuisine') AS cuisine,
    COALESCE(avg_rating::Text, 'No rating') AS avg_rating,
    COALESCE(total_ratings::Text,'No rating') AS total_ratings,
    COALESCE(price_range::Text, 'Nil') AS price_range,
    COALESCE(delivery_available::Text,'Nil') AS delivery_available
   FROM zomato_restuarants ;

select COALESCE(order_id::TEXT, 'No id') AS order_id,
    COALESCE(restaurant_id::TEXT, 'No id') AS restaurant_id,
    COALESCE(customer_id::TEXT, 'No id') AS customer_id,
    COALESCE(order_date::TEXT, 'Nil') AS order_date,
    COALESCE(order_time::TEXT, 'Nil') AS order_time,
    COALESCE(delivery_time::TEXT, 'Nil') AS delivery_time,
    COALESCE(total_cost::Text,'Nil') AS total_cost,
    COALESCE(item_count::Text,'Nil') AS item_count,
    COALESCE(payment_method::Text,'Nil') AS payment_method,
    COALESCE(customer_rating::Text,'No rating') AS customer_rating
FROM zomato_orders;
```

```sql
--Remove duplicate records -
no duplicate records in order_id or restuarant_id
```

Count the number of restaurants in each city.

	city text	total_rest_ids bigint	total_rest_name_count bigint
1	Chennai	91	91
2	Kolkata	92	92
3	Delhi	93	93
4	Bangalore	109	109
5	Mumbai	115	115

Find the top 5 cities with the highest number of orders.

	city text	total_orders bigint
1	Mumbai	3384
2	Bangalore	3300
3	Delhi	2857
4	Kolkata	2732
5	Chennai	2727

Calculate the total revenue generated by each restaurant.

	restaurant_name character varying	total_revenue real
1	Restaurant_116	50931.266
2	Restaurant_262	48860.23
3	Restaurant_56	48645.777
4	Restaurant_27	48562.1
5	Restaurant_127	48203.2
6	Restaurant_141	47153.34
7	Restaurant_208	47065.023
8	Restaurant_396	46774.977
9	Restaurant_48	44596.81

Find the average order amount for each city.

Skill Up. L

	city text	avg_amount double precision
1	Chennai	1041.1271365578025
2	Bangalore	1040.8770784389612
3	Kolkata	1039.502518479332
4	Mumbai	1035.9983008905506
5	Delhi	1033.9865080873826

Identify the top 5 restaurants with the highest total sales.

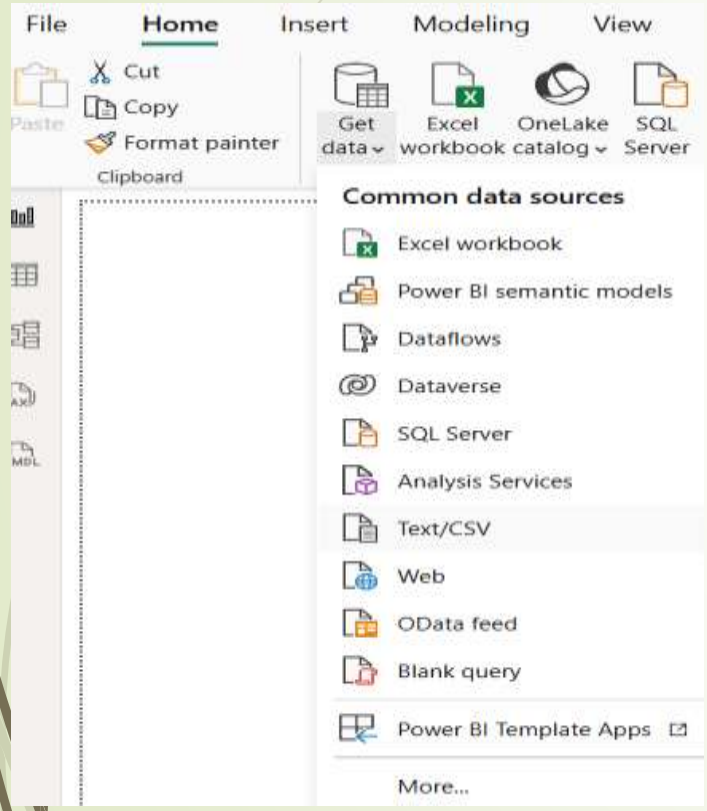
	restaurant_name character varying	total_sales real	total_qty bigint
1	Restaurant_116	50931.266	46
2	Restaurant_262	48860.23	41
3	Restaurant_56	48645.777	44
4	Restaurant_27	48562.1	43
5	Restaurant_127	48203.2	50

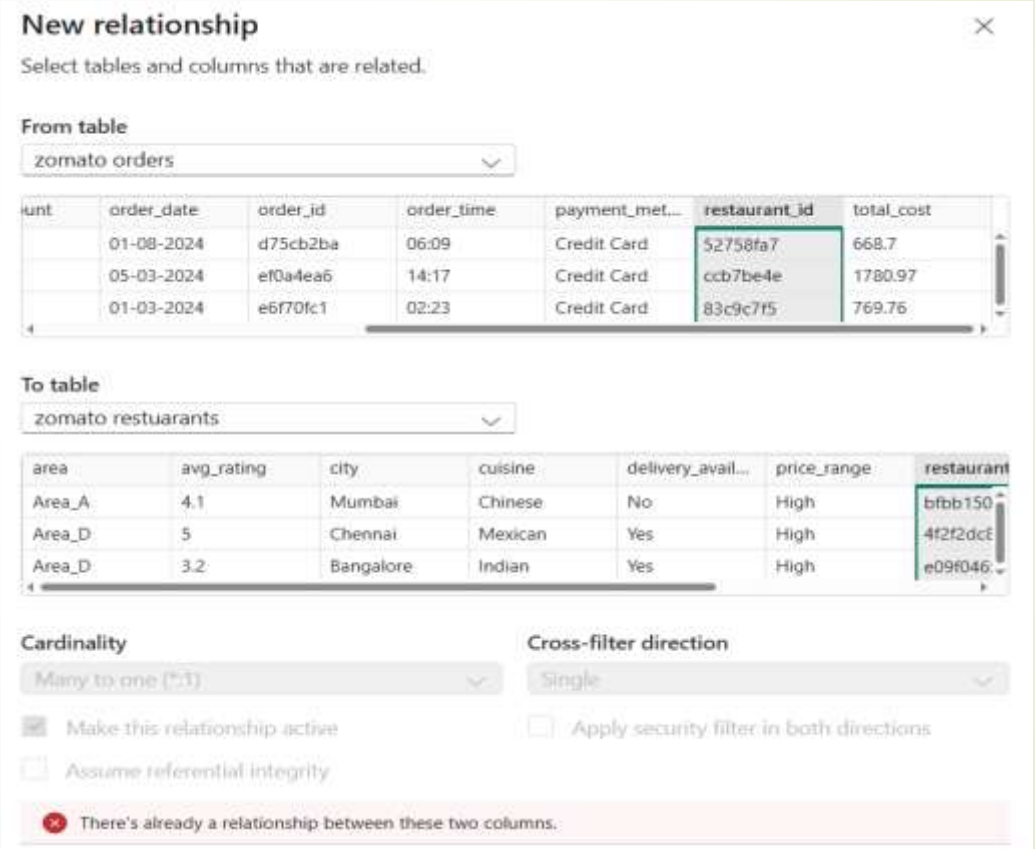
Join the `Zomato_Orders` and `Zomato_Restaurants` tables to get restaurant names along with order details.

Showing rows: 1 to 1000 Page No: 1 of 15

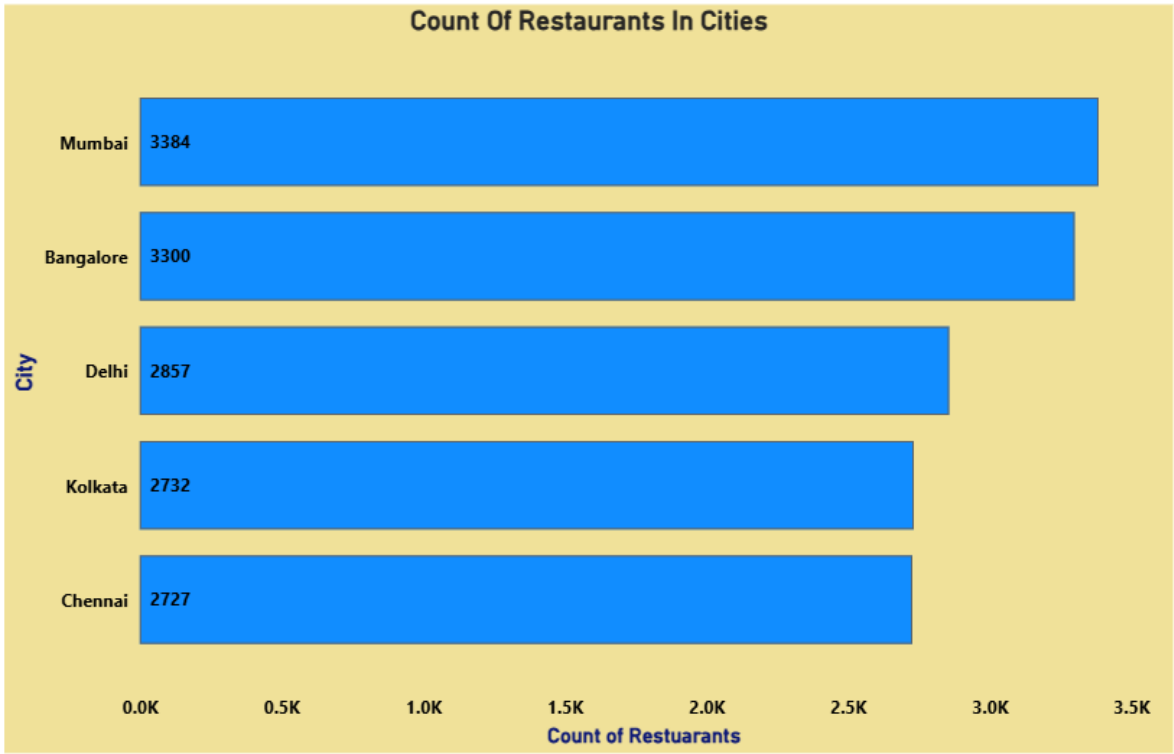
	restaurant_name character varying	order_id character varying	restaurant_id character varying	customer_id character varying	order_date date	order_time time without time zone	delivery_time integer	total_cost real	item_count integer	payment_meth text
1	Restaurant_457	38221dfc	760d1a28	ef2b0995	2024-02-10	16:47:00	79	1344.17	5	UPI
2	Restaurant_343	04b2dc08	502053dd	bad21b62	2024-01-15	17:17:00	63	1672.1	4	Net Banking
3	Restaurant_336	6e1098fe	74d2ed66	308d0b15	2024-05-06	09:51:00	32	992.86	1	Wallet
4	Restaurant_487	d32f543e	d56c5b5e	a9e0d903	2024-02-02	06:29:00	31	1033.8	3	Credit Card
5	Restaurant_326	db1a2c5f	c867f4c3	639c9f46	2024-07-28	02:40:00	50	1365.96	9	Wallet
6	Restaurant_361	f9b9c14d	c2dd51cf	4fc9bd6b	2024-12-04	12:55:00	31	918.36	6	COD
7	Restaurant_458	617e3e23	f5e8e8a9	3c5aacd7	2024-12-08	02:37:00	85	732.02	4	Net Banking
8	Restaurant_263	d09cc1e9	e11f3ddf	bb87ba85	2024-09-14	11:11:00	17	618.44	10	UPI

Task 3 : Power BI Analysis and Visualization



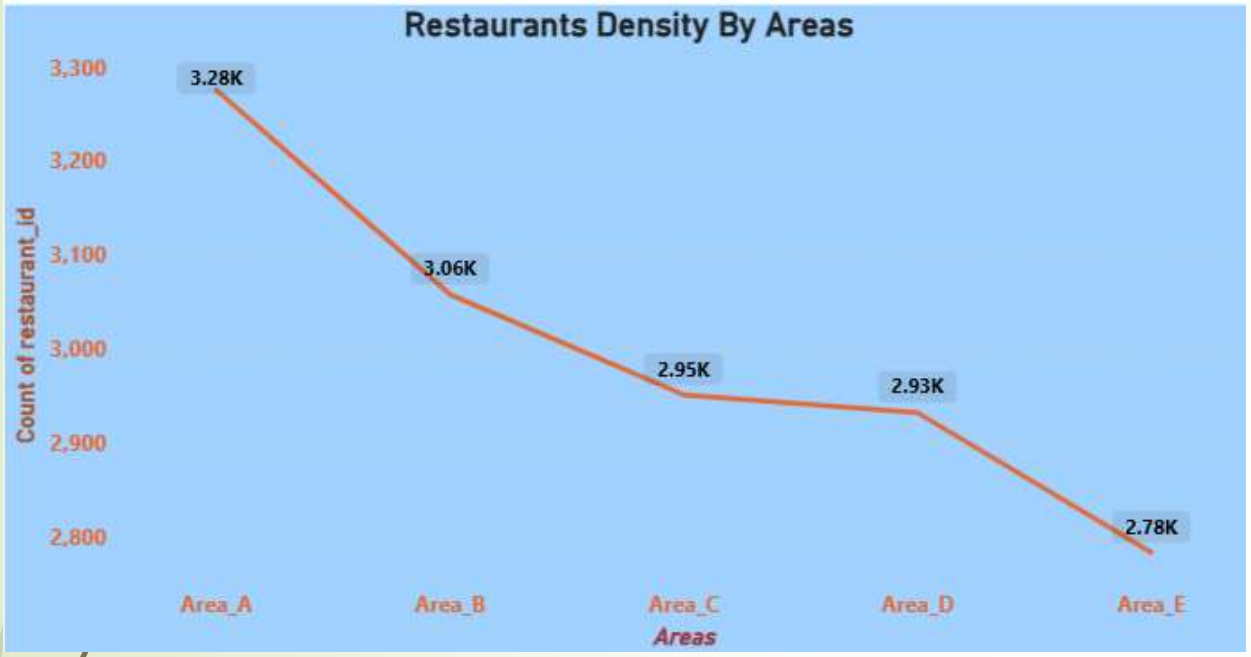


Power BI
Task 1 : Number of Restaurants per City



- At 3,384, Mumbai had the highest Count of restaurants and was 24.09% higher than Chennai, which had the lowest Count of restaurant at 2,727.

Restaurants Areas With High and Low Density



- Area_A had the highest Count of restaurants at 3,276 and was 17.71% higher than Area_E, which had the lowest Count of restaurants at 2,783.

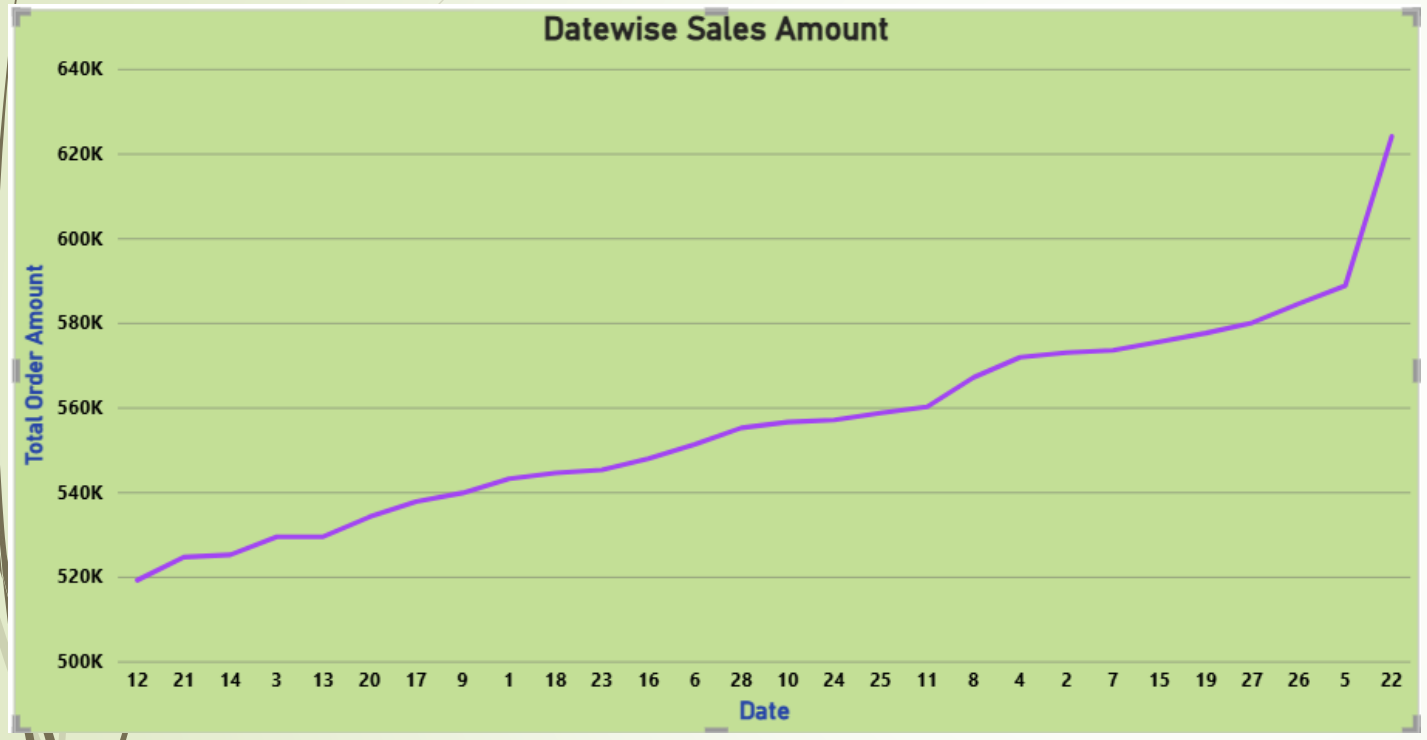
Task 2 : Percentage of Orders from Different Cities

- Mumbai accounted for 22.56% of total count of orders,followed by Banglore for 22%.
- Chennai recorded for least share of orders at 18.18%

Insights with Comparison of Cities in Zomato order market

- Mumbai and Bangalore have both high supply and high demand, meaning those markets are currently the most active and balanced
- Delhi, Kolkata & Chennai have a high density of restaurants (around 90-100) but the order volume is much lower compared to Mumbai and Bangalore.

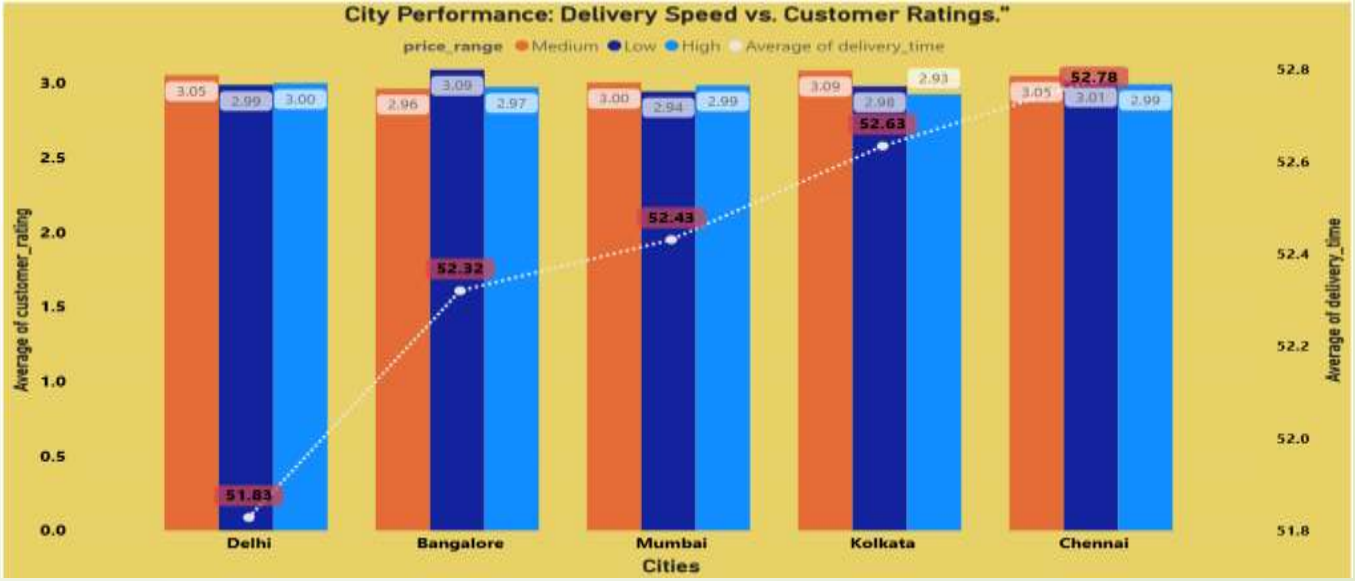
Task 3 : Order Amount Trends Over Time



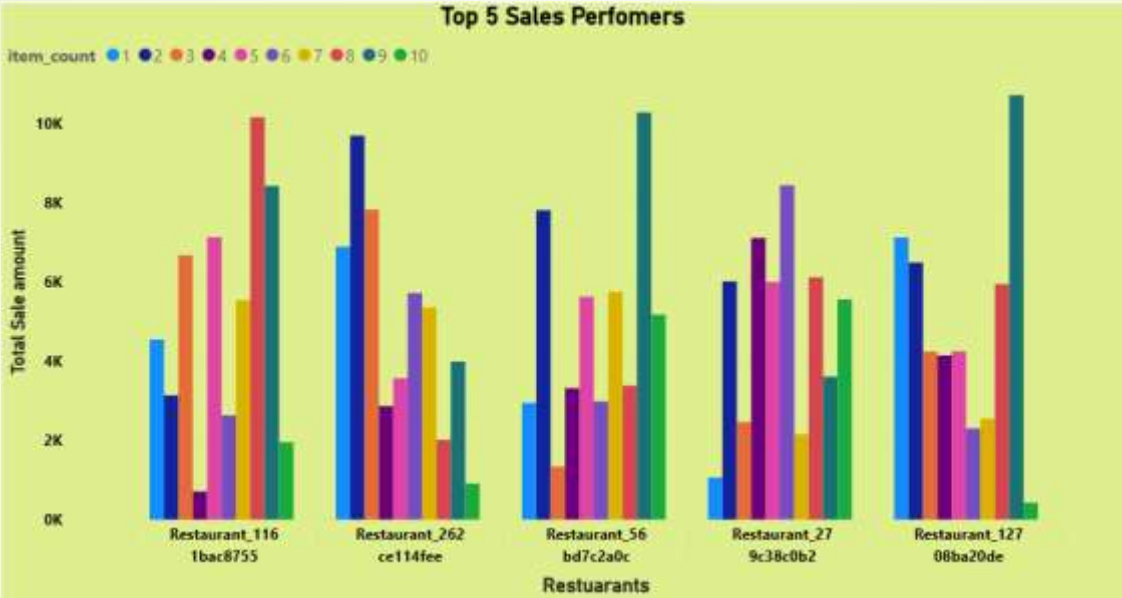
- The date 22nd to 27th shows massive spike in this trend. This can be usually due to festive Seasons: Diwali, Christmas & New Year and weekends during month end.
- Dates from 12th to 21st shows low spending due to mid-week days.

Task 4: Correlation of Factors Affecting Average Rating

- Delhi is the top-performing city in terms of speed. It has the lowest average delivery time (approx. **51.8 minutes**).
- **Medium** and **Low** price ranges frequently outperform the **High** price range in ratings denotes customers paying "High" prices might be more critical, resulting in slightly lower average ratings compared to "Medium" or "Low" tier options.



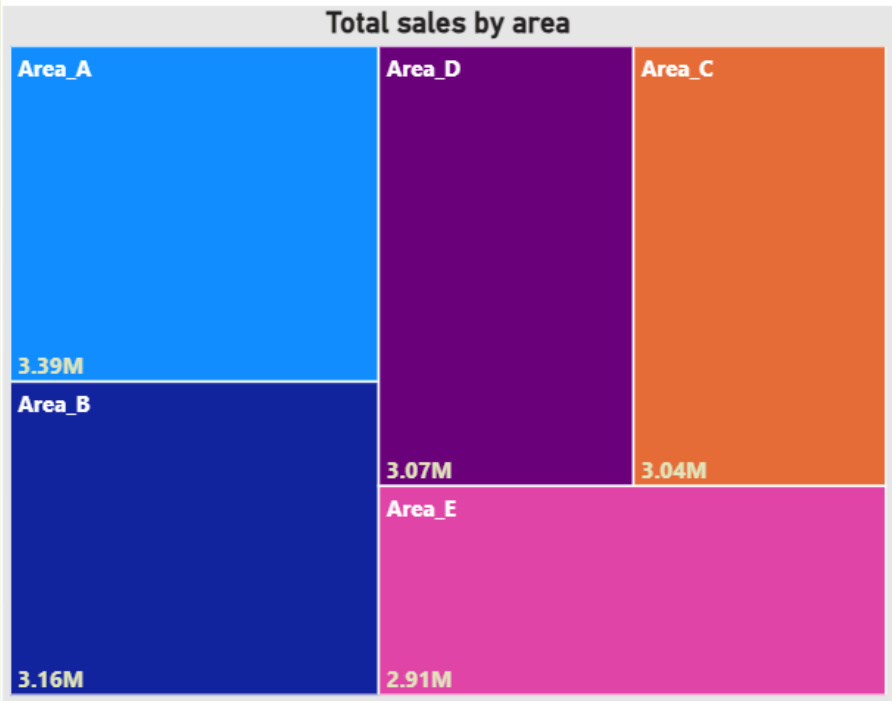
Task 5: Top 5 Restaurants by Total Sales



- These top 5 restaurants are winning through larger order sizes A huge portion of their revenue comes from orders with 6+ items, especially 9–10 items.

- This explains why their total revenue is higher even though average ratings and delivery times are almost the same as everyone else.

Task 6: Revenue by Area Using a Tree Map



- Area_A is the clear leader — it contributes the highest sales and has a noticeable edge over the others.
- The sales are relatively balanced overall — no single area dominates massively, but there is a clear hierarchy: A > B > D > C > E.

Task 7: Order Density by City Using a Heat Map

- Order Density Map shows that Mumbai and Bangalore have the largest bubbles, indicating they have the highest count of orders(most likely Area A & B) representing high density orders.

- The bubbles for Delhi, Kolkata & Chennai on your density map are significantly smaller(likely to be Area C,D & E)

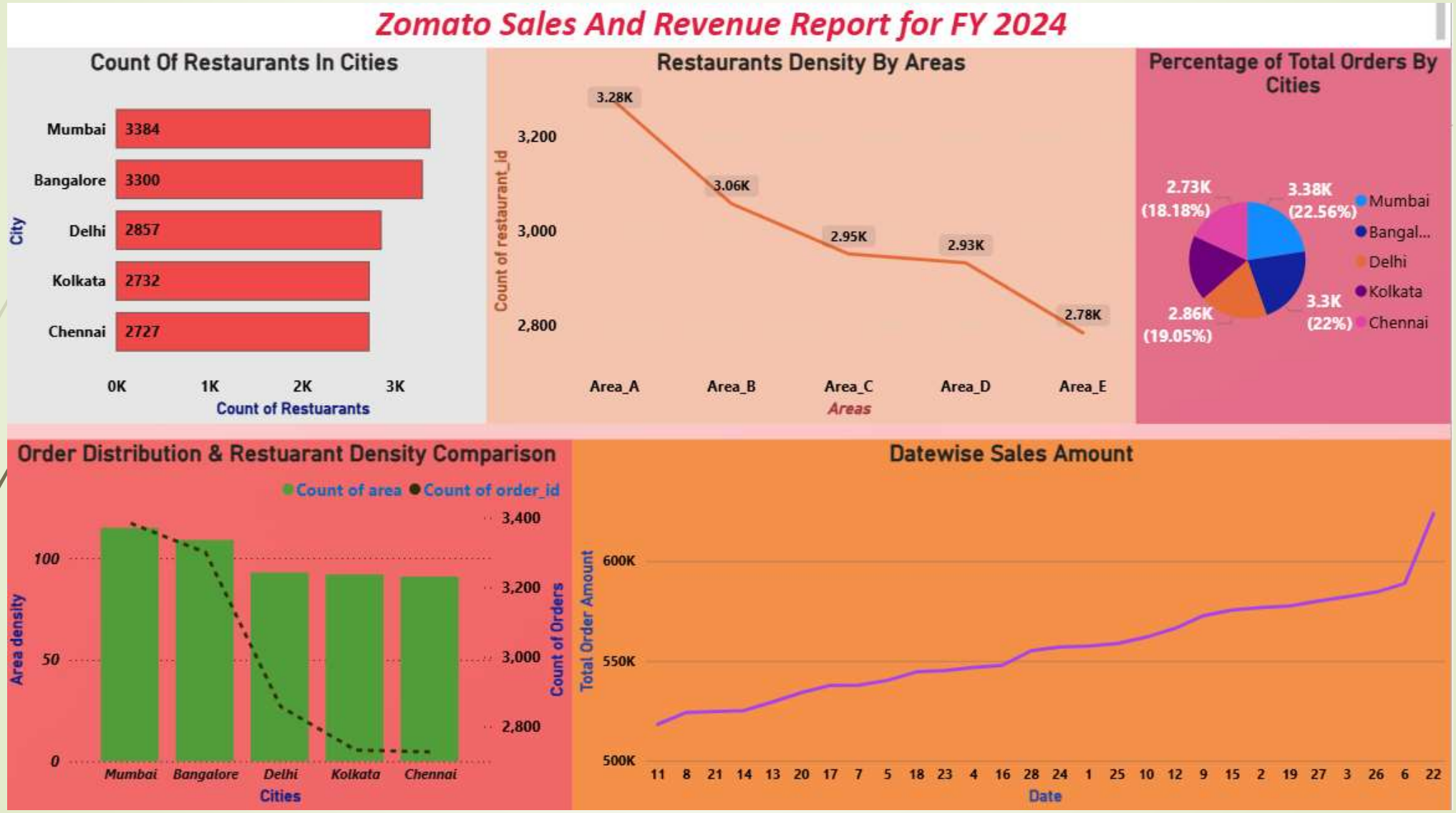
Task 8 : KPI Cards – Total Revenue & Average Order Value

Task 9: Restaurant-Wise Sales Summary in a Table

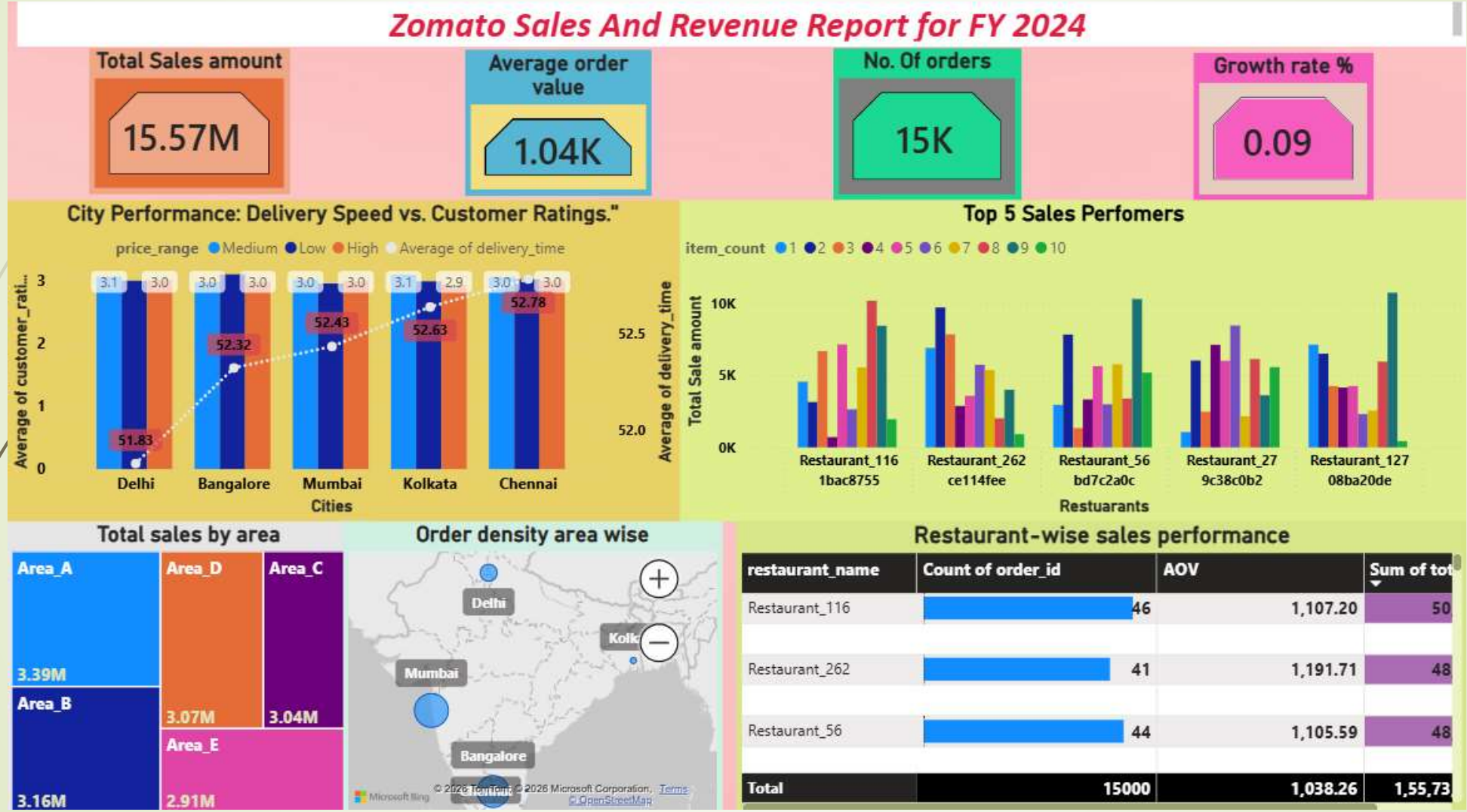
Restaurant-wise sales performance

restaurant_name	Count of order_id	AOV	Sum of total_cost
Restaurant_116	46	1,107.20	50,931.26
Restaurant_262	41	1,191.71	48,860.24
Restaurant_56	44	1,105.59	48,645.78
Restaurant_27	43	1,129.35	48,562.10
Restaurant_127	50	964.06	48,203.20
Restaurant_141	40	1,178.83	47,153.34
Restaurant_208	45	1,045.89	47,065.02
Restaurant_396	40	1,169.37	46,774.97
Restaurant_48	39	1,143.51	44,596.81
Total	15000	1,038.26	1,55,73,886.64

Task : 10 (Dashboard)



Task : 10 (Dashboard)



Key Recommendations

- Focus marketing and promotional efforts on Delhi, Kolkata, and Chennai. While these cities have a high density of restaurants, their current order volumes are significantly lower.
- Implement strategies to improve delivery speeds in cities other than Delhi. Delhi is currently the top performer with the lowest average.
- Prepare for increased staffing and stock between the 22nd and 27th of the month. These periods often coincide with weekends and festive seasons like Diwali, Christmas, and New Year.
- Introduce mid-week special offers or discounts to combat low spending trends typically observed between the 12th and 21st of the month.
- Encourage “mega" deals or family-sized meal options to increase total revenue. Top-performing restaurants generate a massive portion of their revenue from large orders containing 6 to 10 items.